Supporting Information for "Enceladus' and Dione's floating ice shells supported by minimum stress isostasy"

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Contents of this file

1. Text S1 to S7
2. Figures S1 to S5
3. Tables S1 to S7

Introduction

Text S1: Nonhydrostatic deviations of Enceladus and Dione

Text S2: Gravity data: explanation of Tables S2 and S3

Text S3: Shape data: explanation of Tables S4 and S5

Text S4: Basic equations of Bayesian inversion

Text S5: Bayesian inversion results: explanation of Tables S6 and S7

Text S6: Second-order error on the isostatic model: explanation of Figure S4

Text S7: Comment on a paper by *Cadek et al. (2016)*

Figure S1: Prior and posterior distributions of interior parameters for Enceladus

Figure S2: Bayesian confidence regions for core radius and core density

Figure S3: Enceladus' shell thickness if porous crust

Figure S4: Second-order error on the isostatic model

Figure S5: Degree of compensation for top and bottom loading in Enceladus

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Table S1: General parameters for Enceladus and Dione

Table S2: Enceladus' gravity

Table S3: Dione's gravity

Table S4: Enceladus' shape

Table S5: Dione's shape

Table S6: Enceladus: mean values and 1σ (Bayesian) confidence intervals

Table S7: Dione: mean values and 1σ (Bayesian) confidence intervals

Text S1 Nonhydrostatic deviations of Enceladus and Dione

The degree-two shape and gravity ratios are defined by [*Zharkov et al.*, 1985; *Nimmo et al.*, 2011]

$$F_2 = -\frac{H_{20}}{H_{22}} \quad \text{and} \quad G_2 = -\frac{C_{20}}{C_{22}} = \frac{J_2}{C_{22}}\,, \tag{S.1}$$

where H_{nm} and C_{nm} denote the cosine real harmonic coefficients of degree n and order m of the shape and nondimensional gravity potential, respectively. If the body is in hydrostatic equilibrium, $F_2 = G_2 = 10/3$ to first order in the flattening, whatever the density stratification. Second-order corrections change the shape and gravity ratios of a hydrostatic body by up to a few percent with respect to 10/3, with a slight dependence on the internal structure. The degree-two shape and gravity ratio resulting from Eqs. (4)-(5) are

$$F_2 = \frac{10}{3}\left(1 - \frac{48}{35}\,h_2^F\,q\right) \quad \text{and} \quad G_2 = \frac{10}{3}\left(1 - \frac{16}{7}\,h_2^F\,q\right). \tag{S.2}$$

If the body is homogeneous, $(F_2, G_2) = (3.26, 3.21)$ for Enceladus and $(F_2, G_2) = (3.31, 3.30)$ for Dione. Cassini observations yield $(F_2, G_2) = (4.2, 3.5)$ for Enceladus (Tables S2 and S4) and $(F_2, G_2) = (5.2, 4.0)$ for Dione (Tables S3 and S5). To second order in the flattening, the nonhydrostatic shape deviations (with respect to the homogeneous body) are thus $\delta F_2 = 29\%$ for Enceladus and $\delta F_2 = 58\%$ for Dione, while the nonhydrostatic gravity deviations are $\delta G_2 = 9\%$ for Enceladus and $\delta G_2 = 21\%$ for Dione. The density stratification decreases these values by less than one percent ($\delta F_2 = 28\%$ and $\delta G_2 = 8\%$ for Enceladus if $k_2^F = 0.93$).

Text S2 Gravity data (Tables S2 & S3)

Two gravity solutions are available for Enceladus [*Iess et al.*, 2014]. SOL1 includes the five coefficients of degree two and the zonal coefficient of degree three, i.e. the least set

of parameters able to fit the data at the noise level and compatible with the topography. SOL2 includes the five coefficients of degree two and the seven coefficients of degree three. Given the uncertainties, SOL2 does not contain more physical information than SOL1. On the other hand, the errors on SOL1 cannot reflect all the biases due to the sparse data (only three flybys) or resulting from the assumptions involved in spacecraft trajectory modelling (discussed in the Supplementary Material of *Iess et al.* [2014]). In particular, the central value of C_{22} changes from SOL1 to SOL2 by more than 4σ while C_{30} changes by more than 1σ and becomes compatible with a zero value within 2σ (the shift in C_{20} is negligible). The error bars being much larger on SOL2, the shift of C_{22} may be meaningless. Nevertheless we retain both gravity solutions in order to assess the impact of gravity field uncertainties on interior modelling. Table S2 gives the coefficients differing from zero at 3σ (at least for SOL1): C_{20}, C_{22}, and C_{30}. Table S3 gives the coefficients of the preliminary gravity solution for Dione [*Hemingway et al.*, 2016].

The geoid, to first order in the figure of equilibrium, is given by $H^h_{nm} = R_0 C_{nm}$, except for the two components affected by the rotation and by the permanent static tide [*Zharkov*, 2004; *Zharkov and Gudkova*, 2010]:

$$H^h_{20}/R_0 = C_{20} - \frac{5}{6}\,q \quad \text{and} \quad H^h_{22}/R_0 = C_{22} + \frac{1}{4}\,q\,. \tag{S.3}$$

The second-order geoid is given by Eq. (50) of *Zharkov and Gudkova* [2010] but second-order corrections are only a few meters.

Text S3 Shape data (Tables S4 & S5)

The shape of the mid-size Saturnian satellites has been determined by fitting an ellipsoid [*Thomas et al.*, 2007, 2016; *Nadezhdina et al.*, 2016] and by directly estimating the spherical harmonic components of the topography [*Nimmo et al.*, 2011]. For Enceladus and Dione, we use the shape of *Nimmo et al.* [2011] (denoted TOPA), which provides all harmonic coefficients of degree two and three. For Enceladus, we also use the ellipsoidal shape of *Thomas et al.* [2016] (denoted TOPB), which is based on more recent data, from which the harmonic coefficients of degree two can be derived. TOPB is less flattened than TOPA and its coefficient H_{22} follows the geoid (see Table S4). The new ellipsoidal shape of *Nadezhdina et al.* [2016] has a polar flattening ($-3H_{20}/2R$) close to the one of TOPA but its equatorial flattening ($6H_{22}/R$) is nearly hydrostatic (similarly to TOPB).

We use here spherical harmonic coefficients in unnormalized form (as in *Iess et al.* [2014]; *McKinnon* [2015]) so that H_{nm} denote the coefficients of $P_{nm}(\cos\theta)\cos m\phi$ (error bars on H_{nm} are not correct in *Iess et al.* [2014] but are correctly given in *McKinnon* [2015]). *Thomas et al.* [2016] give estimates for the semi-major axes $a > b > c$ and the mean radius R_0 of the best-fitting ellipsoid with 2σ error bars; the mean radius is defined as the radius of the sphere of equal volume; its value in *Thomas et al.* [2016] should be

252.0 km (P. Thomas, private communication). The degree-two harmonic coefficients of the shape, to first order in the flattening, are computed from

$$H_{20} = -\frac{2}{3}\left(\frac{a+b}{2} - c\right) \quad \text{and} \quad H_{22} = \frac{a-b}{6}\,. \tag{S.4}$$

Text S4 Basic equations of Bayesian inversion

The result of the Bayesian inversion is the posterior probability density function, i.e. the conditional probability for the parameters $\mathcal{P}$ given the data $\mathcal{D}$ [*Tarantola*, 2005]:

$$\mathrm{p}(\mathcal{P}|\mathcal{D}) = \frac{\mathcal{L}(\mathcal{D}|\mathcal{P})\,\Pi(\mathcal{P})}{\int d\mathcal{P}\;\mathcal{L}(\mathcal{D}|\mathcal{P})\,\Pi(\mathcal{P})}\,. \tag{S.5}$$

Π represents the prior information on $\mathcal{P}$ while $\mathcal{L}$ is the likelihood function associating a probability to the data given parameter values. $\mathcal{P}$ is related to $\mathcal{D}$ by Eqs. (1) to (5) and (7), though we replace in practice Eqs. (4)-(5) by the complete second-order solutions for the figure of equilibrium. We assume that the gravity data (C_{20}, C_{22}, C_{30}) are normally distributed and uncorrelated (Tables S2 and S3). The model parameters are the densities (ρ_s, ρ_o) and thicknesses (d_s, d_o) of the ice shell and ocean (the radius r_c and density ρ_c of the core are derived parameters). The prior information on $\mathcal{P}$ is described by uniform probability density functions (ranges given in Section 3.4) subjected to the constraint that the shell thickness at the south pole is not negative (Fig. S1 shows the resulting prior distributions).

If the data $\mathcal{D}$ includes the shape coefficients (H_{20}, H_{22}, H_{30}), the isostatic shape coefficients must be treated as additional model parameters. We choose instead to include the shape coefficients with their uncertainties in the model itself which becomes probabilistic. This is similar to the problem of fitting a straight line to (x, y) data with errors in both coordinates. According to Section 4.8.2 of *Gregory* [2005], the likelihood function is the same as if the shape was exactly known except that the standard deviation of the gravity data is replaced by

$$\sigma_{nm}^2 = \sigma_{Cnm}^2 + (Z_n\,\sigma_{Hnm})^2\,, \tag{S.6}$$

where the σ_{Cnm} and σ_{Hnm} are the 1σ uncertainties on the gravity (Tables S2 and S3) and shape (Tables S4 and S5) coefficients, respectively. The likelihood function $\mathcal{L}$ reads

$$\mathcal{L}(\mathcal{D}|\mathcal{P}) = \mathcal{L}_{20}\,\mathcal{L}_{22}\,\mathcal{L}_{30}\,, \tag{S.7}$$

where

$$\mathcal{L}_{nm} = \frac{1}{\sqrt{2\pi}\sigma_{nm}} \exp - \frac{\left(C_{nm}^h + Z_n(H_{nm} - H_{nm}^h) - C_{nm}\right)^2}{2\sigma_{nm}^2}\,, \tag{S.8}$$

it being understood that $H_{30}^h = C_{30}^h = 0$.

We simulate p with a Metropolis-Hastings sampler [*Tarantola*, 2005], generating 400000 samples for each inversion and retaining every fourth sample in order to reduce the correlation within the samples. The burn-in length is equal to 1000. From the selected samples, we compute for each parameter the probability density function (Fig. S1), the mean value and the Bayesian confidence intervals (Tables S6 and S7). The correlation between parameters is illustrated by Fig. 2 showing the two-parameter Bayesian confidence regions for the ice shell and ocean thicknesses, while Fig. S2 does the same for the core radius and density.

Text S5 Bayesian inversion results (Tables S6 & S7)

Table S6 shows the results of several inversions of gravity-shape data for Enceladus, corresponding either to different datasets or to different interior models. SOL1 and SOL2 refer to the gravity data of Table S2 while TOPA and TOPB refer to the shape data of Table S4. The SOL1/TOPA column shows the results of the reference case discussed in the text (Fig. S1 shows the prior and posterior distributions). The SOL2/TOPA column shows the results with the alternative gravity solution SOL2. In that case, the mean value of the shell thickness ($d_s = 24\,\mathrm{km}$) is slightly higher than with SOL1 ($d_s = 23\,\mathrm{km}$) because the larger data uncertainties lead to a distribution extending to much larger values of the shell thickness, though the mode is $d_s = 20\,\mathrm{km}$ (Fig. S1).

The next two columns of Table S6 show the results with different shape data. The shape given by TOPB is only of degree two, thus we compare inversions for SOL1/TOPA and SOL1/TOPB without the third degree. The last column of Table S6 shows the results for an alternative interior model (POROUS) in which the near-surface porosity is a free parameter (see Section 3.4). As expected, the estimated shell thickness is significantly higher because the surface topography contributes less to the gravity coefficients. Fig. S3 shows the dependence of the estimated shell thickness on crustal porosity (for that figure, inversions were made for given values of the porosity).

Table S7 shows the results of the inversion of gravity-shape data for Dione. SOL1 and TOPA refer to the gravity and shape data of Tables S3 and S5. The important effect of the shape uncertainties is illustrated by an inversion in which the shape is precisely known (TOPB).

The distribution of the local shell thickness at the colatitude θ and longitude ϕ is obtained by computing for each sample

$$d_{loc}(\theta,\phi) = d_s + \sum_{n=2,3} H_{n0}^{crust}\, P_{n0}(\cos\theta) + H_{22}^{crust}\, P_{22}(\cos\theta)\, \cos 2\phi\,, \qquad \text{(S.9)}$$

where $H_{nm}^{crust} = H_{nm}^{iso} - H_{nm}^{int}$ and P_{nm} are the unnormalized associated Legendre functions. For Enceladus' model SOL1/TOPA, the parameters in the right-hand side have mean

values and 1σ uncertainties given by $d_s = 22.8 \pm 4\,\mathrm{km}$, $H_{20}^{crust} = -12.1 \pm 2.4\,\mathrm{km}$, $H_{22}^{crust} = 1.3 \pm 0.3\,\mathrm{km}$, and $H_{30}^{crust} = 3.7 \pm 0.7\,\mathrm{km}$. These parameters are correlated so that the error on $d_{loc}(\theta, \phi)$ can only be estimated from the complete distribution.

Text S6 Second-order error on the isostatic model (Figure S4)

In this paper, gravity and shape are split into hydrostatic and isostatic components. Second-order corrections are included in the figure of equilibrium but not in the isostatic model. In order to estimate the error due to this approximation, we compute the second-order gravitational potential C_{nm}^{tot} generated by the total shape (hydrostatic plus isostatic) of all density interfaces (core/ocean, ocean/crust, and surface). The second-order error on the isostatic model is estimated by the difference $C_{nm}^{tot} - (C_{nm}^{h} + C_{nm}^{iso})$.

Consider first a spherically symmetric body of radius R stratified into layers of uniform density. Next, add topographic relief to the interface of radius D between two layers which have a positive density contrast $\Delta\rho$. Powers of the relief can be expanded in real spherical harmonics: $(h/D)^p = s_{inm}^{(p)} Y_{inm}(\theta, \phi)$, where the first subscript denotes cosine ($i = 1$) or sine ($i = 2$) components. The gravitational potential caused by this relief at the surface is given by [*Wieczorek and Phillips*, 1998]

$$U = 4\pi G D^2 \Delta\rho \sum_{i,n,m} \left(\frac{D}{R}\right)^{n+1} \frac{s_{inm}^{\mathrm{FA}}}{2n+1} Y_{inm}(\theta, \phi)\,, \tag{S.10}$$

where the finite amplitude (FA) contribution of the shape reads

$$s_{inm}^{\mathrm{FA}} = \sum_{p=1}^{n+3} \left(\frac{s_{inm}^{(p)}}{p!}\right) \left(\frac{\prod_{j=1}^{p}(n+4-j)}{n+3}\right). \tag{S.11}$$

If the only nonzero shape coefficients are $(s_{20}, s_{22}, s_{30}) = (s_{120}^{(1)}, s_{122}^{(1)}, s_{130}^{(1)})$, the finite amplitude terms associated to these coefficients are given to second order ($p \leq 2$) by

$$\begin{aligned} s_{20}^{\mathrm{FA}} &= s_{20} + \frac{4}{7}\left((s_{20})^2 - 12\,(s_{22})^2 + \frac{2}{3}\,(s_{30})^2\right), \\ s_{22}^{\mathrm{FA}} &= s_{22} - \frac{8}{7}\,s_{20}\,s_{22}\,, \\ s_{30}^{\mathrm{FA}} &= s_{30} + \frac{4}{3}\,s_{20}\,s_{30}\,. \end{aligned} \tag{S.12}$$

For Enceladus, the second-order error on the isostatic model is 5 to 10% of C_{20}^{iso}, 10 to 26% of C_{22}^{iso}, and 10 to 20% of C_{30}^{iso} (Fig. S4a). However, the degree-two isostatic components are a small fraction of the total gravity coefficient (less than 10% of C_{20} and 3% of C_{22} for Enceladus). Therefore the second-order error on the isostatic model is less

than 1% of the degree-two total gravity coefficients and less than 20% of the degree-three gravity coefficient, which is comparable to the uncertainty on the data. Finite amplitude corrections are smaller for Dione.

Text S7 Comment on a paper by *Cadek et al. (2016)*

Outline. A recent paper by *Cadek et al.* [2016] claims that Enceladus' lithosphere provides flexural elastic support so that the shell can be thinner than predicted by classical isostasy. We show here that their claim is wrong.

In that paper, the classical Airy relation between the surface and compensating topography is replaced by one depending on an additional parameter: the elastic thickness T_e of the lithosphere (see Eq. (S.13) below). Because of this new free parameter, the gravity-shape problem becomes underdetermined (see their Fig. 2) and the average shell thickness must be constrained by independent information provided by librations. *Cadek et al.* tune T_e so that the shell thickness matches the one required in the libration model (20 km). Thus it is not surprising that they draw the same conclusions about a thin south polar crust as do the libration models with nonhydrostatic crustal boundaries [*Van Hoolst et al.*, 2016].

We will make two points:

1. The flexural-isostatic model used in *Cadek et al.* [2016] is wrong: it is based on an equation for top loading, but the degree of compensation is computed with a formula valid for bottom loading. If flexural isostasy results from top loading, T_e should be about 20 m instead of 200 m as stated in the paper.

2. Supporting more than 1 km of topography with a 20 m-thick lithosphere stretches belief. More generally, whatever the exact thickness of the lithosphere (as long as it is thin), the degree-two deflection required to provide elastic support induces stresses that lead to lithospheric failure.

We round off the section by discussing the problems raised by isostasy in thin shell theory.

Flexural-isostatic model. In our notation, the flexural-isostatic model of *Cadek et al.* [2016] (their Eq. (1)) reads

$$C_{el}\,\rho_s\,H^{iso}_{nm}\,gR^2 + (\rho_o - \rho_s)\,H^{int}_{nm}\,g_o R_o^2 = 0\,, \tag{S.13}$$

where (R_o, g_o, H^{int}_{nm}) are the radius, gravitational acceleration, and nonhydrostatic topography at the crust-ocean boundary, respectively. The parameter C_{el} is the degree of compensation for elastic flexure which varies between 0.75 and 0.9 in *Cadek et al.* [2016] ($C_{el} = 1$ in the limit of Airy isostasy). With the approximation $gR^2/(g_o R_o^2) \approx 1$, Eq. (S.13) reduces to

$$H^{int}_{nm} = -\frac{\rho_s}{\rho_o - \rho_s}\,C_{el}\,H^{iso}_{nm}\,, \tag{S.14}$$

where H_{nm}^{int} should be interpreted as the deflection of the shell due to the surface topography H_{nm}^{iso}. Eq. (S.14) is the well-known compensation equation for top loading [*Turcotte et al.*, 1981] (without geoid effects), and we expect a choice for C_{el} that is consistent with top loading [*Turcotte et al.*, 1981]:

$$C_{el} = C_{top} = \frac{\chi_n}{\tau\,\phi_n + \chi_n}\,, \tag{S.15}$$

where $\phi_n = n(n+1) - 2$ and $\chi_n = \phi_n + 1 + \nu$ (ν is Poisson's ratio). The bending resistance is neglected here so that there is only one parameter measuring the rigidity of the shell [*Turcotte et al.*, 1981]:

$$\tau = \frac{ET_e}{R^2 g(\rho_o - \rho_s)}\,, \tag{S.16}$$

where E is Young's modulus and T_e is the elastic thickness of the lithosphere.

Instead, *Cadek et al.* define C_{el} by a formula which was proposed for the modelling of dynamic topography (Eq. (25) of *Kalousová et al.* [2012]), a kind of bottom loading:

$$C_{el} = C_{bot} = \frac{\chi_n}{\tau'\,\phi_n + \chi_n}\,, \tag{S.17}$$

where τ' is defined by

$$\tau' = \frac{ET_e}{R^2 g \rho_o}\,. \tag{S.18}$$

The correct formula for the shell deflection due to bottom loading would be (Eq. (A10) of *McGovern et al.* [2002]):

$$H_{nm}^{int} = -\frac{\rho_o}{\rho_o - \rho_s}\,\frac{1}{C_{bot}}\,H_{nm}^{iso}\,, \tag{S.19}$$

where H_{nm}^{int} should be interpreted as the thickness of the bottom load and H_{nm}^{iso} as the shell deflection. Clearly, there is a contradiction in using Eq. (S.14) with C_{el} defined by Eq. (S.17). The correct way to proceed would be to use Eq. (S.14) with the degree of compensation for top loading, Eq. (S.15).

If $\rho_o = 1030\,\mathrm{kg/m^3}$ and $\rho_s = 925\,\mathrm{kg/m^3}$, $\tau' \approx \tau/10$ and C_{bot} is much larger than C_{top} (Fig. S5). In particular, $C_{bot} = 0.84$ corresponds to $T_e = 200\,\mathrm{m}$, which is the elastic thickness favoured in *Cadek et al.* [2016]. By contrast, using the correct formula for the degree of compensation ($C_{top} = 0.84$) yields $T_e = 20\,\mathrm{m}$. It is hardly credible that more than 1 km of topography can stand on such a thin lithosphere without breaking it (see below).

Flexural isostasy with top loading is not the only possibility. Alternatively, Eqs. (S.17) and (S.19) can be used if the deflection of the shell (or crust) results from a subsurface density anomaly. Beware that the thickness of the bottom load is not identical to the topography of the crust-ocean boundary: the former is measured with respect to the deflected crust-ocean boundary whereas the latter is measured with respect to the spherical surface coinciding with the crust-ocean boundary before deflection. If top and bottom loads are

both present, the flexure problem cannot be solved without specifying another free parameter, which is the ratio of subsurface to surface loading (Eq. (B5) of *McGovern et al.* [2002]).

Lithospheric failure. If we put aside the question of how C_{el} is related to T_e, the flexural-isostatic model of *Cadek et al.* [2016] (Eq. (S.13)) can be seen as a parametric relation between surface topography and bottom topography, in which C_{el} is a black-box parameter representing the effect of flexural isostasy. Following *McKinnon* [2013], we now show that a thin lithosphere (less than 200 m thick) cannot support degree-two topography without breaking up, except in the limit of full isostasy with negligible deflection.

The degree-two zonal deformation of a spherical shell generates tangential membrane stresses $(\sigma_{\theta\theta}, \sigma_{\phi\phi})$ given by the Vening-Meinesz equations (Eqs. (4.2)-(4.3) of *Melosh* [2011]). We quantify the stress in the lithosphere by the maximum differential stress

$$\Delta\sigma = \mathrm{Max}(\sigma_{\theta\theta} - \sigma_{\phi\phi}) \approx \mu|\Delta f|\,, \tag{S.20}$$

where $\mu = 3.5\,\mathrm{GPa}$ is the crustal shear modulus and Δf is the shell flattening. The latter is related to the $(2,0)$ harmonic component of the shell deflection (H_{20}^{flex}) by $\Delta f = (-3/2)H_{20}^{flex}/R$. We neglect here the equatorial flattening because Enceladus' nonhydrostatic deformation is mostly zonal.

If compensation results from top loading, H_{20}^{flex} is equal to the deflection of the shell given by H_{20}^{int} in Eq. (S.13), which is much larger in magnitude than the topographic load itself because of the near-complete compensation ($C_{el} = 0.84$). The hydrostatic-isostatic decomposition yields $H_{20}^{iso} \approx -1300\,\mathrm{m}$ so that the maximum differential stress is $\Delta\sigma \approx 240\,\mathrm{MPa}$. If compensation results from bottom loading, H_{20}^{flex} is equal to the shell deflection (H_{20}^{iso} in Eq. (S.19)), so that the maximum differential stress is $\Delta\sigma \approx 30\,\mathrm{MPa}$. Both estimates are much larger than the tensional strength of intact cold ice (1 to 2 MPa) [*Schulson and Duval*, 2009]. The lithosphere actually fails at lower stress levels because of its finite brittle strength, as discussed in detail in *McKinnon* [2013]. In conclusion, the nonhydrostatic degree-two topography on Enceladus cannot be supported by lithospheric flexure. The only way out consists in distributing the loads on the top and bottom of the crust so as to minimize the shell deflection and the resulting membrane stresses. But this state of full isostasy does not depend on the elastic thickness of the lithosphere.

Isostasy in thin shell theory. Applying thin shell theory to isostasy is problematic for three reasons. First, thin shell theory is not valid if the shell is thicker than ten percent of the shell radius. Thus one must suppose a priori that the ice shells of Enceladus and Dione are less than 25 km and 56 km thick, respectively. Enceladus's ice shell seems to satisfy, but just barely, the thin shell threshold whereas Dione's ice shell does not. Thin shell theory could be valid for Dione (or Enceladus) if the lithosphere is much thinner than the whole ice shell. It is not obvious, however, how the subsurface loading applied on the crust-ocean boundary is transmitted to the lithosphere itself (see third reason below). Second, the

geoid cannot be included in a consistent way: the topographic loads are measured with respect to the geoid which is affected, in turn, by the deflection of the shell as well as by the surface and subsurface loads themselves. This effect is crucial at long wavelengths. Third, the standard (first-order) formalism of thin shell theory does not take into account the difference in area between the inner and outer shell surfaces. It is thus not possible to introduce ad hoc factors gR^2 and $g_o R_o^2$ as was done by *Cadek et al.* (compare Eqs. (S.13) and (S.14)), since factors of similar magnitude are neglected elsewhere in thin shell theory. This problem must be solved by working with a higher-order approximation of thin shell theory which is not worth the effort in comparison with the minimum stress isostasy developed in our paper.

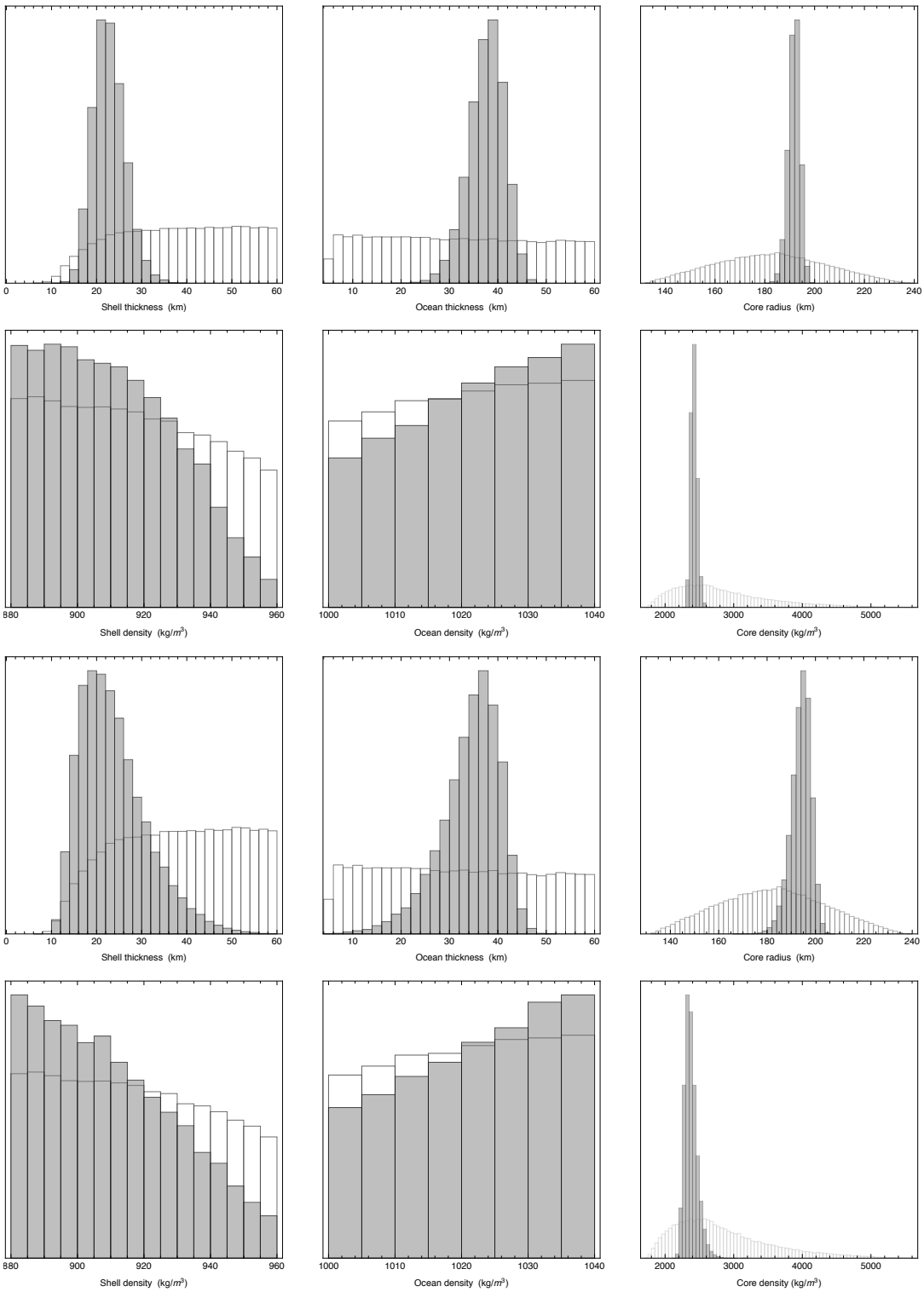


Figure S1: Prior and posterior distributions of interior parameters for Enceladus: thickness and density of the crust (or ice shell), ocean, and core. The models are SOL1/TOPA (top six panels) and SOL2/TOPA (bottom six panels). Priors are transparent and posteriors are shaded. Priors are the same in the two models.

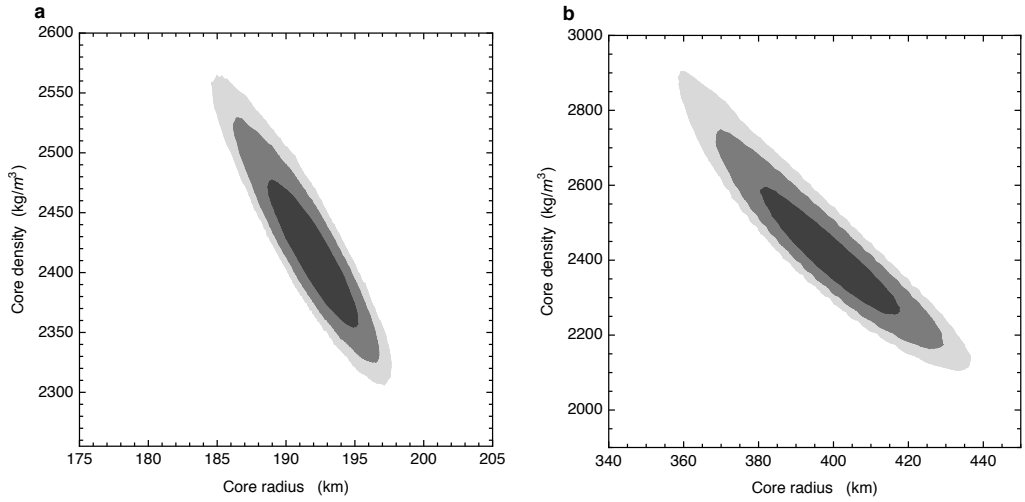


Figure S2: Bayesian confidence regions for core radius and core density. (a) Enceladus, (b) Dione. Contours show Bayesian confidence regions to $(1\sigma, 2\sigma, 3\sigma)$ for the model SOL1/TOPA. The inverse correlation between the core radius and core density is clearly visible.

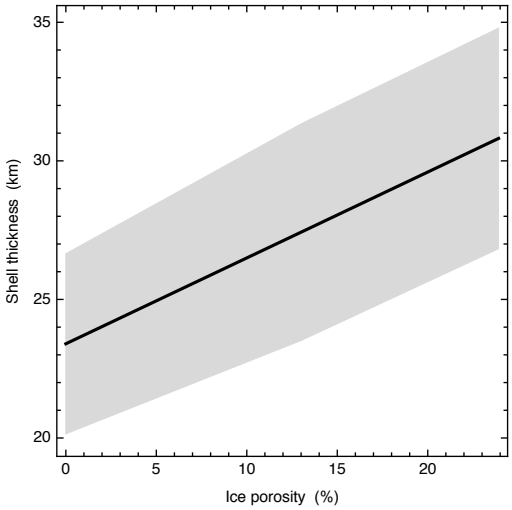


Figure S3: Enceladus' shell thickness if porous crust. The crust is split in a bottom layer with a density of 920 kg/m^3 and a 10 km-thick upper layer with a porosity varying between 0% (920 kg/m^3) and 24% (700 kg/m^3). The solid curve shows the mean value resulting from the Bayesian inversion while the shaded area shows the 1σ confidence interval. Several inversions were made for given values of the porosity, contrary to the POROUS model of Table S6 in which near-surface porosity is a free parameter.

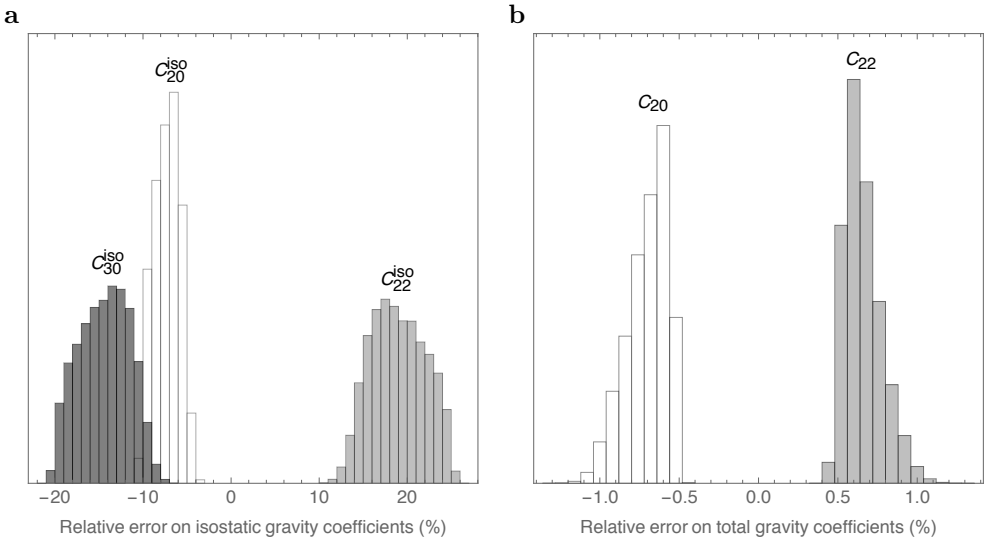


Figure S4: Second-order error on the isostatic model. Distribution of the relative error on (a) the isostatic gravity coefficients, (b) the total gravity coefficients. The satellite is Enceladus and the model is SOL1/TOPA.

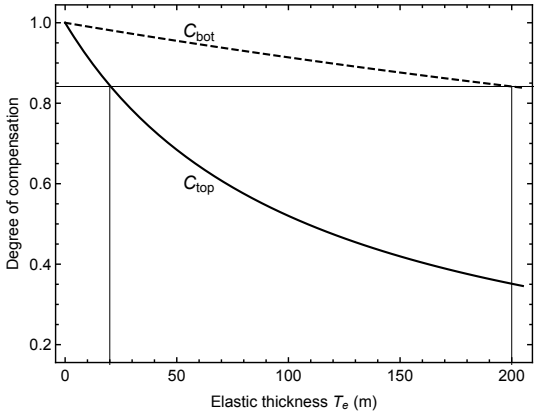


Figure S5: Degree of compensation for top and bottom loading in Enceladus. The factors C_{top} (top loading, solid curve) and C_{bot} (bottom loading, dashed curve) are given by Eqs. (S.15) and (S.17), respectively. The straight lines indicate the T_e values corresponding to $C_{top} = 0.84$ and $C_{bot} = 0.84$. The densities are $(\rho_o, \rho_s) = (1030, 925)\,\mathrm{kg/m^3}$ and the elastic parameters are $(E, \nu) = (9.3\,\mathrm{GPa}, 0.33)$.

Table S1: General parameters for Enceladus and Dione. Surface radii are taken from Tables S4 and S5. Gravitational parameters are taken from SOL1 of *Iess et al.* [2014] for Enceladus and from JPL online data (http://ssd.jpl.nasa.gov/) for Dione. The bulk density and surface gravity are computed from R and GM (Tables S2–S3). The rotational parameter is defined by $q = \omega^2 R^3/GM$ with $\omega = 2\pi/T$.

Parameter	Symbol	Enceladus	Dione	Unit
Surface radius	R	252.1	561.4	km
Gravitational parameter	GM	7.21044	73.1146	$\mathrm{km}^3/\mathrm{s}^2$
Bulk density	$\bar{\rho}$	1610	1478	$\mathrm{kg/m}^3$
Surface gravity	g	0.113	0.232	$\mathrm{m\,s}^{-2}$
Period	T	1.37	2.737	days
Rotational parameter	q	6.26×10^{-3}	1.71×10^{-3}	-

Table S2: Enceladus' gravity. The nondimensional gravity coefficients C_{nm} [*Iess et al.*, 2014] are rescaled from the gravity field reference radius (254.2 km) to $R = 252.1$ km as in *McKinnon* [2015]. Error bars are 1σ. ε_1 and ε_2 are the relative errors (in %) for SOL1 and SOL2, respectively. See Text S2.

Coeff.	SOL1 ($\times 10^6$)	ε_1	SOL2 ($\times 10^6$)	ε_2
C_{20}	-5526 ± 35	0.6	-5534 ± 70	1
C_{22}	1576 ± 16	1	1640 ± 53	3
C_{30}	118 ± 23	20	88 ± 72	80
G_2	3.51 ± 0.04	1	3.37 ± 0.12	4

Table S3: Dione's gravity. SOL1 is the preliminary solution given in *Hemingway et al.* [2016]. Same notation as in Table S2.

Coeff.	SOL1 ($\times 10^6$)	ε_1
C_{20}	-1454 ± 16	1
C_{22}	363 ± 2	0.6
G_2	4.01 ± 0.05	1

Table S4: Enceladus' shape. TOPA and TOPB are the shape solutions of *Nimmo et al.* [2011] and *Thomas et al.* [2016], respectively, expressed in terms of unnormalized real spherical coefficients H_{nm}. R_0 is the radius of the sphere of equal volume. As in the original papers, error bars are 2σ for (a, b, c, R_0, F) and 1σ for (H_{nm}, F_2). ε_A and ε_B are the relative errors (in %) on the harmonic coefficients of TOPA and TOPB, respectively. GEO1 and GEO2 are the geoids computed to first order with the gravity solutions SOL1 and SOL2, respectively. See Text S3.

	TOPA	ϵ_A	TOPB	ϵ_B	GEO1	GEO2	Unit
a	$256.8 \pm 0.2^*$		256.2 ± 0.3				km
b	$251.3 \pm 0.2^*$		251.4 ± 0.2				km
c	$248.3 \pm 0.4^*$		248.6 ± 0.2				km
R_0	252.1 ± 0.2		252.0 ± 0.2				km
H_{20}	-3846 ± 179	5	$-3470 \pm 90^{**}$	3	-2708 ± 9	-2710 ± 18	m
H_{22}	917 ± 19	2	$800 \pm 30^{**}$	4	792 ± 4	808 ± 13	m
H_{30}	384 ± 5	1	-	-	30 ± 6	22 ± 18	m
F_2	4.20 ± 0.21	5	4.33 ± 0.20	5	-	-	-

* Error bars on the differences $(a - R_0, b - R_0, c - R_0)$.

** Error bars assuming that errors on (a, b, c) add quadratically.

Table S5: Dione's shape. See Table S4 for details.

	TOPA	ϵ_A	GEO1	Unit
a	$563.5 \pm 0.3^*$			km
b	$561.3 \pm 0.3^*$			km
c	$559.5 \pm 0.3^*$			km
R_0	561.4 ± 0.3			km
H_{20}	-1923 ± 157	8	-1616 ± 9	m
H_{22}	368 ± 32	9	444 ± 1	m
F_2	5.2 ± 0.6	12	-	-

* Error bars on $(a - R_0, b - R_0, c - R_0)$.

Table S6: Enceladus: mean values and 1σ (Bayesian) confidence intervals. The inverted parameters are the shell thickness d_s, ocean thickness d_o, shell density d_s, and ocean density d_o. The core radius r_c and core density ρ_c are derived parameters without prior. The prior on ρ_s is not valid for the POROUS model (see Section 3.4). The symbol $*$ indicates that the parameter is not constrained by the inversion. The numbers $(nm, ...)$ in the second row denote the degree n and order m of the harmonic coefficients used in the inversion. The layer thicknesses sum to $R \pm 1\,\mathrm{km}$ because the results are rounded to the nearest integer. See Text S5.

Parameter	Prior range	SOL1/TOPA $(20, 22, 30)$	SOL2/TOPA $(20, 22, 30)$	SOL1/TOPA $(20, 22)$	SOL1/TOPB $(20, 22)$	POROUS $(20, 22, 30)$
d_s (km)	$1 - 60$	23 ± 4	24 ± 7	31 ± 9	31 ± 9	27 ± 4
d_o (km)	$5 - 60$	38 ± 4	34 ± 6	31 ± 8	31 ± 8	29 ± 5
r_c (km)	–	192 ± 2	194 ± 4	190 ± 3	190 ± 3	196 ± 3
ρ_s (kg/m^3)	$880 - 960$	$*$	$*$	$*$	$*$	$*$
ρ_o (kg/m^3)	$1000 - 1040$	$*$	$*$	$*$	$*$	$*$
ρ_c (kg/m^3)	–	2422 ± 46	2377 ± 94	2477 ± 76	2479 ± 76	2370 ± 44

Table S7: Dione: mean values and 1σ (Bayesian) confidence intervals. See Table S6 for the notation. TOPB is the same as TOPA but with zero error bars. The layer thicknesses sum to 562 km because the results are rounded to the nearest integer. See Text S5.

Parameter	Prior range	SOL1/TOPA	SOL1/TOPB
d_s (km)	$1 - 200$	99 ± 23	102 ± 10
d_o (km)	$5 - 200$	65 ± 30	57 ± 16
r_c (km)	–	398 ± 14	403 ± 9
ρ_s (kg/m^3)	$880 - 960$	$*$	$*$
ρ_o (kg/m^3)	$1000 - 1040$	$*$	$*$
ρ_c (kg/m^3)	–	2435 ± 140	2383 ± 58